Worksheet 7.1

How big is an atom?

Student: ..... Class: .....

It is hard to visualise the size of tiny objects, such as atoms and their nuclei. However, making comparisons with something we are more familiar with sometimes helps.

1. Use an everyday object to create an image that will help people to understand how small the nucleus of an atom is compared with the size of the entire atom. Choose from the suggestions provided or use your own ideas. Atoms vary in size, but for the purpose of this activity assume the nucleus of an atom is 100 000 times smaller than the size of the entire atom.

(a) If a marshmallow represents the nucleus of an atom, how big would the entire atom be? (*Hint:* The length of a marshmallow is approximately 2 cm.)

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(b) If one gradation on millimetre-lined graph paper represents the nucleus of an atom, how many sheets of graph paper would you need to glue together to represent the entire atom?

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(c) If a marble, about 1 cm in diameter, representing a nucleus is placed in the centre of a sportsground, how big would the ground need to be to represent the entire atom?

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Photo: Josh Casey / 123RF.com

Building with atoms

Imagine going to hospital to have an incredibly tiny machine injected into your body! The machine could whiz through your body to heal a damaged knee or to fight a disease in your liver. A machine like this might be a thing of the future, but research into nanotechnology is well underway.

Nanotechnology is the science of using single atoms as building materials. Currently, scientists have been able to use nanotechnology to build tiny machines, some of which have moving parts.

2. Imagine if a tiny machine (nanobot) was required to swim through a particular blood vessel, measuring 1 cm in diameter. The size of the machine would be limited by many factors, including the number of atoms that could comfortably fit inside the blood vessel. How many atoms could fit side-by-side in this blood vessel? (*Hint:* Find out how big a typical atom is.)

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